

## 📌 Interview Questions (Clear Answers)

### 1. Tree vs Forest?

Decision Tree: Single model, prone to overfitting

Random Forest: Collection of trees → more accurate & stable

### 2. Gini Impurity vs Information Gain?

Gini Impurity: Measures how mixed classes are

Information Gain: Based on entropy (reduction in randomness)

👉 Both are used to decide splits

### 3. How Random Forest handles overfitting?

Uses multiple trees

Each tree trained on different data (sampling)

Averages predictions → reduces variance

### 4. What is Bagging?

Bootstrap Aggregation

Random sampling with replacement

Train multiple models → combine results